Representation of Data

ANSWER KEY

Section 1: Stem-and-Leaf Diagrams & Box Plots

1 In an ordered stem-and-leaf diagram of 15 test scores, the key states $3 \mid 5 = 35$. The leaves for stem 4 are $1, 3, 5, 8, 8$. What is the median score if there are 4 scores below stem 4?

a. **45** ✓

b. 43

c. 48

d. 41

For $n = 15$, the median is the 8th value: 4 values are in stem 3, so in stem 4 we count the 5th, 6th, 7th, 8th values (1, 3, 5, 8). The 8th value is 45.

2 A data set has minimum $= 12$, $Q_1 = 20$, median $= 28$, $Q_3 = 36$, and maximum $= 52$. What is the interquartile range (IQR)?

a. **16** ✓

b. 24

c. 40

d. 8

$\text{IQR} = Q_3 - Q_1 = 36 - 20 = 16$.

3 Using the $1.5 \times \text{IQR}$ rule on the data from the previous question, determine if the maximum value 52 is an outlier.

a. **No, the upper outlier boundary is 60** ✓

b. Yes, any value above 48 is an outlier

c. Yes, any value above 50 is an outlier

d. No, because the range is less than 50

$\text{Boundary} = Q_3 + 1.5 \times \text{IQR} = 36 + 1.5(16) = 36 + 24 = 60$. Since $52 \leq 60$, it is not an outlier.

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Section 2: Histograms & Frequency Density

4 In a histogram, the class interval $20 \leq x < 35$ has a frequency of 45. What is the frequency density for this bar?

a. **3.0** ✓

b. 1.5

c. 4.5

d. 2.25

Class width $= 35 - 20 = 15$. Frequency density $= \frac{\text{Frequency}}{\text{Class width}} = \frac{45}{15} = 3.0$.

5 On a histogram, a bar representing the interval $10 \leq x < 20$ has a width of 2 cm and a height of 5 cm, representing 30 items. What area represents 1 item?

a. $\frac{1}{3}$ cm^2 ✓

b. $\frac{1}{2}$ cm^2

c. 3 cm^2

d. 1 cm^2

Area of bar $= 2 \times 5 = 10$ cm^2. Since this represents 30 items, each item is represented by $10/30 = 1/3$ cm^2.

6 Why is frequency density, rather than raw frequency, plotted on the vertical axis of a histogram for grouped continuous data?

a. **Because class widths may be unequal and area must be proportional to frequency** ✓

b. Because continuous data cannot be plotted with integer frequencies

c. To ensure the total height of all bars equals 100

d. To eliminate the need for class boundaries

When classes have unequal widths, plotting frequency density ensures that the area of each rectangle is directly proportional to the frequency of that class.

Section 3: Cumulative Frequency Curves & Percentiles

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7 When plotting a cumulative frequency graph for grouped continuous data, at which points should the cumulative frequencies be plotted?

a. **At the upper class boundaries ✓**

b. At the midpoints of each class

c. At the lower class boundaries

d. At the class frequencies

Cumulative frequency represents the total number of observations up to that value, so it must be plotted at the upper boundary of each class interval.

8 From a cumulative frequency curve of 200 students, the 80th percentile corresponds to which cumulative frequency?

a. **160 ✓**

b. 80

c. 120

d. 180

The 80th percentile corresponds to $0.80 \times 200 = 160$ on the vertical cumulative frequency axis.

9 Using linear interpolation on grouped data, the median lies in the class $40 \le x < 50$ with frequency 25 and cumulative frequency before this class equal to 35. If total $n = 100$, find the estimated median.

a. 46.0 ✓

b. 45.0

c. 44.5

d. 47.2

Median position $= 100/2 = 50$. We need $50 - 35 = 15$ more items out of 25. Median $= 40 + \frac{15}{25}(10) = 40 + 6 = 46.0$.

Section 4: Linear Coding & Transformations of Summary Stats

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10 A data set x has mean $\bar{x} = 50$ and variance $s_x^2 = 16$. A new variable is defined as $y = 3x - 10$. Find the mean $\bar{y}$ and standard deviation s_y.

a. $\bar{y} = 140, s_y = 12$ ✓

b. $\bar{y} = 140, s_y = 36$

c. $\bar{y} = 150, s_y = 12$

d. $\bar{y} = 140, s_y = 4$

Mean: $\bar{y} = 3(50) - 10 = 140$. Standard deviation: $s_y = |a|s_x = 3(\sqrt{16}) = 3(4) = 12$.

11 A variable t is coded as $u = \dfrac{t - 200}{10}$. For 50 observations, $\sum u = 85$ and $\sum u^2 = 320$. Find the actual mean of t.

a. 217 ✓

b. 215

c. 208.5

d. 220

Mean of u: $\bar{u} = 85/50 = 1.7$. Actual mean $\bar{t} = 10\bar{u} + 200 = 10(1.7) + 200 = 217$.

12 For the data in the previous question, calculate the variance of the original variable t.

a. 351 ✓

b. 35.1

c. 320

d. 500

$\text{Var}(u) = \frac{\sum u^2}{n} - \bar{u}^2 = \frac{320}{50} - (1.7)^2 = 6.4 - 2.89 = 3.51$. Then
$\text{Var}(t) = 10^2 \times \text{Var}(u) = 100 \times 3.51 = 351$.

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